

ECE4132 assignment 5: control of a cart-pendulum system

Prepared by: Dr. James Saunderson

Aims

The aim of this assignment is to gain familiarity with the design and implementation of controllers using state space methods.

References and resources

- K. J. Åström and R. M. Murray *Feedback systems: an introduction for scientists and engineers* Princeton University Press
- ECE4132 weeks 7–8 lecture notes and slides.

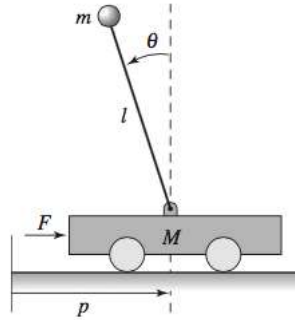
Overview

In this assignment you will

- Investigate the open-loop behaviour of a cart-pendulum system.
- Design a state-feedback based controller for a linearisation of the system.
- Investigate how your controller works on the actual cart-pendulum system.

1 System model

The system we work with in this assignment consists of an inverted pendulum on a cart:



The angular displacement of the pendulum from the vertical position at time t is $\theta(t)$. The position of the cart from a reference position at time t is $p(t)$. The input to the system at time t is a force $F(t)$ acting on the cart as shown.

These variables are related by the nonlinear differential equation

$$\begin{bmatrix} (M + m) & -ml \cos(\theta(t)) \\ -ml \cos(\theta(t)) & (J + ml^2) \end{bmatrix} \begin{bmatrix} \frac{dp}{dt} \\ \frac{d\theta}{dt} \end{bmatrix} = \begin{bmatrix} F(t) \\ 0 \end{bmatrix} - \begin{bmatrix} c \frac{dp}{dt} + ml \sin(\theta(t)) \left(\frac{d\theta}{dt} \right)^2 \\ \gamma \frac{d\theta}{dt} - mgl \sin(\theta(t)) \end{bmatrix}.$$

The parameters of the differential equation are given below:

- Mass of pendulum: $m = 0.2$ kg
- Mass of cart: $M = 0.5$ kg
- Acceleration due to gravity $g = 9.8$ ms⁻²

- Length from pivot to center of mass: $l = 0.3$ m
- Moment of inertia about the center of mass: $J = 0.006$ kgm²
- Coefficient of friction of the cart $c = 0.1$
- Coefficient of friction of the pendulum $\gamma = 0.005$

Throughout, we make the choice of state

$$\mathbf{x}(t) = \begin{bmatrix} p(t) \\ \theta(t) \\ \frac{dp}{dt} \\ \frac{d\theta}{dt} \end{bmatrix}.$$

Using these variables, the system can be expressed in nonlinear state-space form as

$$\frac{d\mathbf{x}}{dt} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & M+m & -ml \cos(x_2(t)) \\ 0 & 0 & -ml \cos(x_2(t)) & J+ml^2 \end{bmatrix}^{-1} \begin{bmatrix} x_3(t) \\ x_4(t) \\ F(t) - cx_3(t) - ml \sin(x_2(t))(x_3(t))^2 \\ -\gamma x_4(t) + mgl \sin(x_2(t)) \end{bmatrix}$$

The system has an equilibrium point at $F = 0$ and $\mathbf{x} = \mathbf{0}$, corresponding to the cart being stationary at position zero, the pendulum being stationary and balancing in the upright vertical position, and there being no external force acting on the cart. If we linearise about the equilibrium point, the resulting linear system is

$$\frac{d\mathbf{x}}{dt} = \mathbf{A}\mathbf{x}(t) + \mathbf{B}F(t)$$

where

$$\mathbf{A} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & M+m & -ml \\ 0 & 0 & -ml & J+ml^2 \end{bmatrix}^{-1} \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & -c & 0 \\ 0 & mgl & 0 & -\gamma \end{bmatrix} \text{ and } \mathbf{B} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & M+m & -ml \\ 0 & 0 & -ml & J+ml^2 \end{bmatrix}^{-1} \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \end{bmatrix}. \quad (1)$$

In different places in the assignment we will consider two different output equations for this model.

Output equal to the full state: Consider the linearised cart-pendulum system:

$$\begin{aligned} \frac{d\mathbf{x}}{dt} &= \mathbf{A}\mathbf{x}(t) + \mathbf{B}F(t) \\ \mathbf{y}(t) &= \mathbf{C}\mathbf{x}(t) + \mathbf{D}F(t) \end{aligned} \quad (2)$$

where the input is the force $F(t)$ on the cart and the output $\mathbf{y}(t)$ is equal to the full state of the system at time t .

Q1 The matrix $\mathbf{C}$ in (2) can be generated by the MATLAB code `eye(m,n)` and the matrix $\mathbf{D}$ in (2) can be generated by the MATLAB code `zeros(p,q)`. What are the values of m , n , p , and q ?

Output equal to the position variable: Consider the linearised cart-pendulum system:

$$\begin{aligned} \frac{d\mathbf{x}}{dt} &= \mathbf{A}\mathbf{x}(t) + \mathbf{B}F(t) \\ y_p(t) &= \mathbf{C}_p\mathbf{x}(t) + \mathbf{D}_pF(t) \end{aligned} \quad (3)$$

where the input is the force $F(t)$ on the cart and the output $y_p(t) = p(t)$ is equal to the position of the cart at time t .

Q2 The matrix $\mathbf{C}_p$ in (3) can be generated by the MATLAB code `eye(m,n)` and the matrix $\mathbf{D}_p$ in (3) can be generated by the MATLAB code `zeros(p,q)`. What are the values of m , n , p , and q ?

2 Getting started

- Download the folder `ECE4132assignment5.zip` and unzip to a location on your MATLAB path.
- Open the script: `setupModelParameters.m` which is where you will be writing your MATLAB code.
- Open the Simulink model `cartPendulum.slx` which is where you will simulate your controllers on the cart-pendulum system.
- Add a constant block to the `cartPendulum.slx` model so that the input force is set to zero.
- In `setupModelParameters.m`, set the initial condition `xinit` to different values, simulate the system, and see what happens. In particular, you should try to following situations:
 - initial angle from the upward vertical is 0, initial angular velocity is 0, initial cart position is 0, and initial cart velocity is 0.1.
 - initial angle from the upward vertical is 0.1, initial angular velocity is 0, initial cart position is 0, and initial cart velocity is 0.

In each case, try to predict what will happen, and then see if the simulated system does what you expect by looking at the system animation. Also take a look at the scope and figure out which state is which on the scope plot.

3 Investigating the linearised system

We now consider the linearised system.

- In the file `setupModelParameters.m`, add code to define the matrices **A** and **B** and **C** and **D** of the state space model of the linearised system.
- Find the eigenvalues of the linearised system. (*Hint*: use the MATLAB command `eig`.)

Q3 Is the linearised system internally stable?

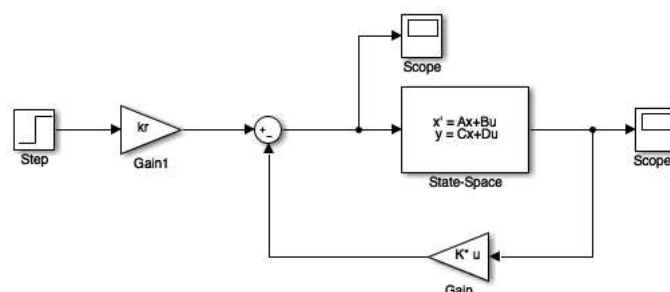
- Add code to `setupModelParameters.m` to define an additional 1×4 matrix **C_p** so that $p(t) = \mathbf{C}_p \mathbf{x}(t)$, i.e., the matrix from Q2.

Q4 Find the poles and zeros of the transfer function from F to p . (*Hint*: use `zpkdata`.) Give your answers correct to two significant figures and order them from smallest to largest.

- Add code to `setupModelParameters.m` to compute the reachability matrix of the linearised system.

Q5 What is the determinant of the reachability matrix? Is the linearised system reachable?

4 Control of the linearised model via state feedback



In this section, you will add a state feedback controller of the following form to your Simulink model of the linearised system:

$$F(t) = -\mathbf{K}\mathbf{x}(t) + k_r r(t)$$

where $r(t)$ is the reference input at time t , $\mathbf{x}(t)$ is the state at time t , and $F(t)$ is the input force acting on the cart at time t .

- Open the Simulink model `linCartPendulum.slx` which is where you will prototype your designs on the linearised model of the system. Check that the model parameters of the state space block are the matrices A, B, C, and D defined in `setupModelParameters.m`.
- Add the feedback controller to the Simulink model `linCartPendulum.slx`. Set the parameters of the gain blocks so that they refer to the values of variables called K and `kr` in your workspace. Choose the correct settings for the gain block so that it implements matrix multiplication.
- In `setupModelParameters.m`, add code that computes a state feedback gain matrix K so that the closed-loop poles of the system are at:

$$-4 \pm j, \quad -6 \pm 0.5j$$

(Hint: use the command `place`.)

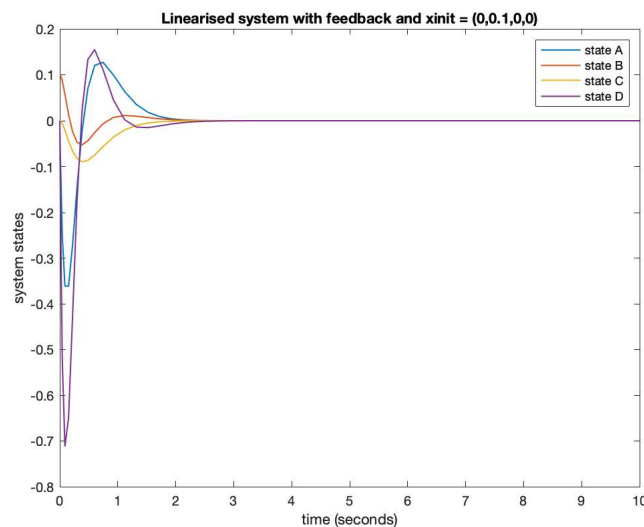
Q6 What value of the matrix K of feedback gains achieves this aim?

- In `setupModelParameters.m`, add code that computes a precompensation gain `kr` so that the if the reference input is constant, then the steady-state value of the cart position is equal to the reference input.

Q7 What value of `kr` achieves this aim?

- **Stabilising:** Set the initial state to `xinit = [0;0.1;0;0]` (i.e., initial position, velocity and angular velocity zero, initial angle 0.1 radians). Set the reference to a constant value of 0. Simulate the feedback system.

Q8 The following plot shows how the four states of the system evolve over time, but the states are labelled A, B, C, D. Match the labels A, B, C, D, to the state variables (position, angular position, linear velocity, and angular velocity).



- **Position control:** Set the initial state to `xinit = [0;0;0;0]`. Set the reference input to be a step from position 0 to position 1 at time $t = 0$. Simulate the feedback system.

Q9 What are the largest (most positive) and smallest (most negative) values of force that the controller exerts on the cart? Give your answers correct to 2 significant figures.

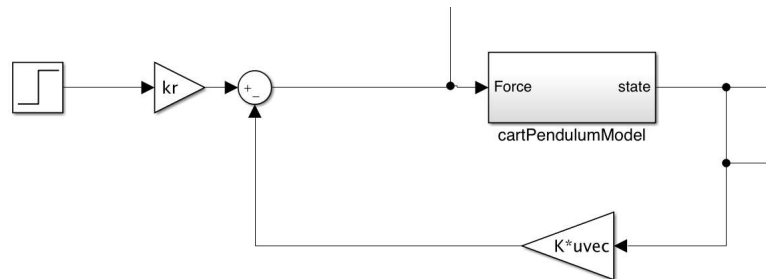
Q10 Find the 2% settling time of the step response of the cart position. Given your answer correct to 2 significant figures. (Hint: use `stepinfo`)

- **Fast position control:** Modify the desired closed loop eigenvalues, and hence K and `kr`, so that the settling time of the step response of the cart position is at most half of what it was before (i.e., the system dynamics are about twice as fast).

- Rerun your simulation, and check that the system achieves a 2% settling time that is at most half of what you found in question 7. If necessary, modify your desired eigenvalue locations until it does achieve the desired specifications.

Q11 Complete the following statement: The largest value of the force that the controller exerts on the cart (increases/decreases/stays about the same) when the feedback gains are modified to reduce the 2% settling time by a factor of two.

5 Control of the nonlinear system via state feedback



- Open the Simulink model `cartPendulum.slx`.
- Add a state feedback controller of the form $F(t) = -\mathbf{K}\mathbf{x}(t) + k_r r(t)$ to the system. Ensure the gain blocks refer to the variables `K` and `kr` in your workspace.
- In `setupModelParameters.m`, ensure that the feedback gains `K` and precompensation gain `kr` are set the same as in Q6 and Q7 from Section 4.
- **Stabilising:** Set the initial state to `xinit = [0;0.1;0;0]` (i.e., initial position, linear velocity, and angular velocity equal to 0; initial angular position equal to 0.1 radians.) Set the reference to a constant value of 0. Simulate the feedback system. Observe what happens.
- Repeat the previous experiment with different values of the initial angular position.

Q12 Which of the following statements is true about the state feedback controller using the design parameters from Q6 and Q7:

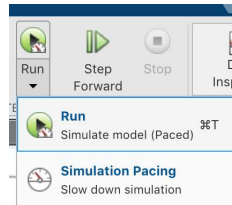
- A The state feedback controller can stabilise the nonlinear system at the equilibrium point $\mathbf{x} = (0, 0, 0, 0)$ if the initial angular position is less than 1. The state feedback controller can stabilise the linearised system at the equilibrium point $\mathbf{x} = (0, 0, 0, 0)$ for any initial value of the angular position.
- B The state feedback controller can stabilise the nonlinear system at the equilibrium point $\mathbf{x} = (0, 0, 0, 0)$ if the initial angular position is less than 0.5 and can stabilise the linearised system at the equilibrium point $\mathbf{x} = (0, 0, 0, 0)$ for any initial value of the angular position.
- C The state feedback controller can stabilise the nonlinear system at the equilibrium point $\mathbf{x} = (0, 0, 0, 0)$ if the initial angular position is less than 0.5, and can stabilise the linearised system at the equilibrium point $\mathbf{x} = (0, 0, 0, 0)$ only if the initial value of the angular position is less than 0.1.
- D The state feedback controller can stabilise the nonlinear system at the equilibrium point $\mathbf{x} = (0, 0, 0, 0)$ if the initial angular position is less than 1, and can stabilise the linearised system at the equilibrium point $\mathbf{x} = (0, 0, 0, 0)$ only if the initial value of the angular position is less than 0.5.

- **Position control:** Set the initial state to `xinit = [0;0;0;0]`. Set the reference input to be a step from position 0 to position 1 at time $t = 0$. Simulate the feedback system.

Q13 What are the largest (most positive) and smallest (most negative) values of force that the controller exerts on the cart? Give your answers correct to 2 significant figures.

Q14 Find the 2% settling time of the step response of the cart position. Given your answer correct to 2 significant figures. (*Hint: use `stepinfo`*)

- **Fast position control:** In `setupModelParameters.m`, change `K` and `kr` to be the same as you found in the item marked 'Fast position control' in Section 4, with the aim of reducing the settling time of the system by a factor of two. Rerun the simulation. If necessary, slow down the simulation to run at something like 0.1 times real-time speed, to see what is happening. You can do this by going to 'simulation pacing' under the 'run' button:



Q15 Which of the following statements is correct about the system response.

- A The controller can move the cart from position 0 to position 1, keep the pendulum balanced, and do so with a 2% settling time that is at most half of what was found in Q13.
- B The controller can move the cart from position 0 to position 1, keep the pendulum balanced, but does not do so with a 2% settling time that is at most half of what was found in Q13.
- C The controller cannot move the cart from position 0 to position 1 while keeping the pendulum balanced.